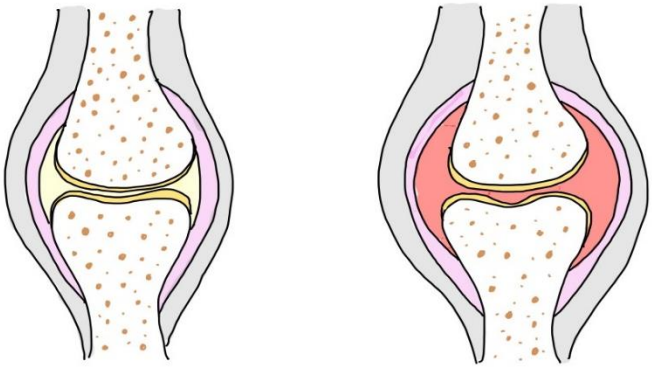
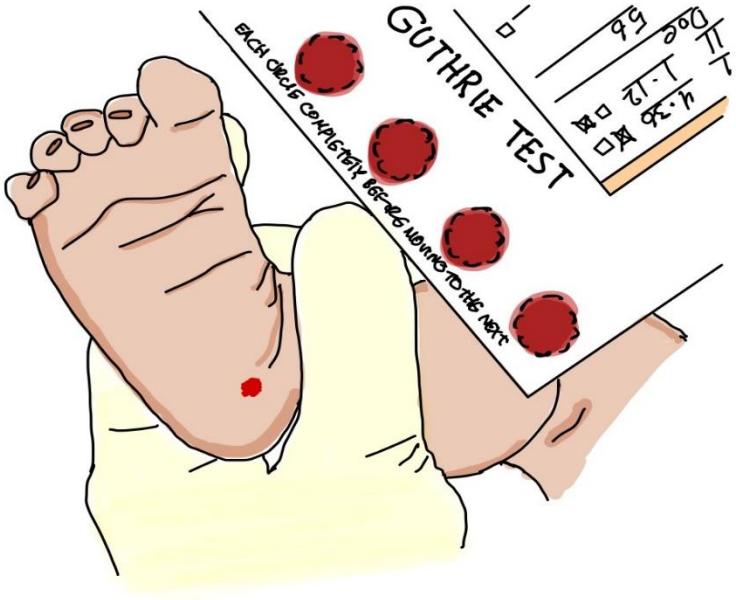
PAEDIATRICS

Rheumatology, Inborn error of metabolism, Poisoning and Immunology





د. عبدالرحمن الجعكي

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Bones, Joints and rheumatic disorders

Variations of normal Posture:

Variations of musculoskeletal normality and differential diagnosis:

Perceived Disorder	Normal Age range	Differential diagnoses to consider
Bow Legs	1–3 years	Rickets – osteogenesis imperfecta, Blounl's disease
Knock Knees	2–7 Years	Juvenile Idiopathic arthritis
Flat Feet (per planus)	1–2 Years	Hypermobility, congenital tarsal fusion
In – Toeing	1–2 years	Tibia torsion, femoral anteversion
Out – Toeing	6–12 months	Hypermobility, Ehlers– Danlos and Marfan's syndromes
Toe walking	1–3 Years	Spastic dipiegia, muscular dystrophy

Disorders of hip joint:

Limping –can be painful and painless

1. Developmental Dysplasia of the hip (DDH):
   - Discussed in Neonatology
2. Transient synovitis (Irritable Hip):
   - This is the most common cause of acute hip pain in children
   - Age :2–12 years
   - Often follow URTI
   - C/P:
     1. Sudden onset of hip pain or limp and may referred to knee
     2. No pain at rest
     3. No fever and pt does not appear ill
   - Ix :
     1. CBC CRP: Normal
     2. X Ray hip: Normal
     3. BL C&S: Normal
   - Rx –
     1. Bed rest
     2. Analgesia
   - Prognosis: Improves within days
3. Perthes disease:
   - Due to ischemia of femoral epiphysis,avascular necrosis followed by revascularization and reossification over 18 –36 months
   - Sex: more in ♂ (5:1) ♂:♀
   - Age :5–10 years of age
   - C/P :Gradual onset of limp or hip pain and Bilateral 10 –20%
   - X Ray shows increase density in female head ,Fragmented and irregular
   - MRI may be needed to confirm Rx

- Rx: Bed rest –skin traction
- Prognosis: In most children the prognosis is good especially if Rx <6 years and <50% of femoral head affected

4. Slipped Upper Femoral Epiphysis:
    - Displacement of the epiphysis of femoral head post–inferiorly
    - Age :10 –15 years
    - More in ♂ , obese, child with hypothyroidism or hypopituitarism
    - C/P: Limping or hip pain (>Trauma) Restriction of abduction and IR
    - 20% Bilateral
    - Rx: By X Ray hip
    - Rx: Usually surgical (Pin fixation)
    - Complication: avascular necrosis

"Osteomyelitis"

- Infection of the metaphysis of long bones
- Site: Distal femur and proximal tibia (most common)
- Pathogen: **Staph aureus (most common)** , Streptococcus and Hib, Salmonella (in SCA)
- Route of transmission
    1. Haematogenous spread
    2. Direct spread from infected wound
- C/P:
    1. Fever
    2. Painful , Immobile limb
    3. Swelling and tenderness
    4. In infant may insidious – swelling with decrease movement
- Investigation
    1. CBC: high WBC(PMN)
    2. CRP: high
    3. Blood C &S:+ ve (usually)
    4. X Ray:Normal early (1st 7–10 days)
    5. Radionuclide bone scan (Diagnostic) Early or MRI .
- Rx
    1. Analgesia
    2. Bed rest – splin
    3. IV Antibiotic (Immediately given) for several 4–6 weeks
    4. Surgical Drainage may needed if not response

DD Acute Onset Limb:

1. Trauma MC
2. Osteomyelitis
3. Bone Tumor
4. Septic Arthritis

"Arthritis"

Causes of Polyarthritis:

Infection	**Bacterial** –septicaemia / septic arthritis TB **Viral** –rubella/ Mumps, adenovirus, coxsackie B, herpes, hepatitis, Parvovirus **Other** – mycoplasma, Lyme disease, rickettsia Reactive– gastrointestinal infection. Streptococcal infection Rheumatic fever
Inflammatory bowel disease	Crohn's Disease, ulcerative colitis
Vasculitis	Henoch– schonlein purpura, kawasaki's disease
Haematological Disorders	Haemophilia , sickle cell disease
Malignant disorders	Leukaemia, neuroblastoma
Connective tissue disorders	Juvenile Idiopathic arthritis (JIA), Systemic lupus erythematosus (SLE), dermatomyositis,Polyarteritis nodosa (PAN)
Other	Cystic Fibrosis

1. "Reactive arthritis"
   - Is the most common form of arthritis in childhood
   - C/P transient joint swelling (<6 weeks) following extra articular infection (salmonella– campylobacter mycoplasma – Lyme or viral infection – STD)
   - CBC:X Ray :Normal
   - Rx : No Rx or NSAID
2. Rheumatic fever and post strep reactive arthritis:
   - Rare in developed countries and common in developing countries

"Septic Arthritis":

- It is serious infection of the joint space lead to bone destruction
- Age :< 2 years of age (any age)
- Route of transmission: Haematogenous spread or Direct spread
- Usually one joint is affected "Hip"
- Organism: **Staph aureus (most common)** and Hib
- Immune decrease and SCA: high Risk
- C/P:
  1. Acute Erythematous worm swollen joint
  2. Decrease range of movement (cry if it is moved) (pseudo paresis)
  3. Fever high grade ,irritability.
- Investigation
  1. CBC: high WBC(PMN)
  2. CRP: high
  3. U/S joint :effusion

4. X Ray :Excluded trauma
5. Bone scan and MRI
6. Aspiration of the joint space :C & S –>Diagnostic

- **Treatment**
  1. Bed rest
  2. Analgesia
  3. IV antibiotic Flueloxacillin for 4–6wks
  4. Surgical drainage of joint

Septic arthritis	Transient synovitis
• ill appearing • Acute onset of pain and high grade fever > 38.5c • CRP: high • Wbc > 12.000cells / ml • Inability to bear weight • Restricted in extreme rotation and refuses to move joint • Rx: Antibiotic & surgical drainge	• Well appearing • Mild or no fever • CRP:normal or slightly raised • Wbc:normal or slightly elevated • Able to bear weight • restricted abduction • Recent URTI • Rx: analgesics & bed rest " self – limiting

- **Henoch – Schonlein purpura:**
  - ✓ Most common vasculitis of childhood
  - ✓ C/P purpuric rash – arthritis – Abdomen pain – haematuria proteinuria
- **Systemic lupus erythematosus: "SLE"**
  - ✓ SLE rare in childhood
  - ✓ More common in adolescent ♀

The Painful Knee:

1. **Osgood – schlatter disease:**
   - Osteochondritis of patellar tendon insertion
   - Increase in adolescent males – football player
   - C/P knee pain > exercise – localized tenderness and sometimes swelling over the tibial tuberosity BiL 25 – 50% – resolve with decrease activity and physiotherapy for Quadriceps MS, knee immobilizer splint may be helpful
2. **Chondromalacia Patellae:**
   - Softening of articular cartilage of patella
   - Increase in adolescent females
   - C/P knee pain on walking up stair or on standing ass e^{-} hypermobility
   - Rx physiotherapy of Quadriceps MS
3. **Osteochondritis dissecans**
4. **Knee Trauma**

Red flag clinical features of back pain:

1. Young Age
2. High fever (infection)
3. Night waking – persistent pain (osteoid osteoma)
4. Painful scoliosis (infection – malignancy)
5. Focal Brain signs loss of power / bladder control – spinal compression
6. Ass e⁻ weight loss – systemic malaise (malignancy)

Growing pains = Nocturnal Idiopathic pain

Is common in preschool and school age group

Criteria:

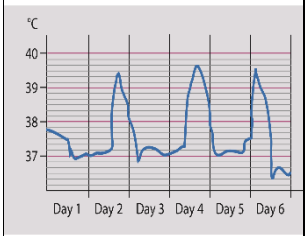
1. Age 3–12 years
2. Pains symmetrical in lower limbs and not limited to joints
3. Pain never present at start of the day after walking
4. Physical activity not limited – No Limb
5. Physical examination normal (including PGALS) with the exception of joint hypermobility in some cases and otherwise well.

Hypermobility:

1. Pain in older children adolescents in lower limbs worse after exercise – No joint swelling or transient
2. Hypermobility can be generalized or limited to peripheral J such hands and feet
3. While mild degrees of hypermobility are normal on young ♀ but may features of some (chromosomal syndrome e⁻ g Down) (Collage disease eg marfan or Ehlers – Danlos syndrome)

Juvenile Idiopathic Arthritis (JIA)

- Is the mc chronic inflammatory disease in children Adolescent in UK
- Diagnostic Criteria
  - Age at onset < 16yrs
  - Arthritis in ≥ 1 joint
  - Duration ≥ 6 wks
  - Exclusion of other causes of arthritis (infection or any other)
- Prevalence: 1:1000 children
- (At least 7 types of JIA) based on the number of joints affected in the 1st 6 months

	Oligoarthritis	Polyarthritis	Systemic Arthritis
%	≃ 50	≃ 20%	≃10%
Nob of Joints	1–4	≥5	oligo or Poly
Type of Joints	Large(Knee,ankle and Wrist) 	Small: fingers or Large joints – TMJ) 	May have pains in joints and muscles (arthralgia/myalgia) but initially no arthritis
Sex	increase ♀ 5:1	increase ♀ 5:1	♀=♂ 1:1
System failures (Extra articular)	• Chronic Uveitis 20% • Asymmetrical growth	• Chronic Uveitis 5% Fever low grade Decrease growth	• Fever spiking – Malaise • Salmon rash • LN – HSM • Serositis 
ANA	+/–	- +/ –	-
Lab Finding	–	-	• Hb low – high WBC • High PLt – high CRP

Types	1. Persistent 49% 2. Extended 8% (>4 Joints)	1. Rh –ve 16% 2. Rh +ve 3% (Rh nodules ≃ 10%)	
Prognosis	1. Excellent 2. Moderate	1. Moderate 2. Poor	Variable to poor

- Psoriatic Arthritis ≃ 7%:age 1–16 year – ♀=♂ 1:1 Large and small J – Psoriasis – Uveitis 20% – nail pitting with Moderate prognosis
- Enthositis related arthritis ≃ 7%:age 6–16 year – increase ♂ 4:1 Large J – Lumbar and sacroiliac – inf of Tendon and Ligament – arthritis HLA B27 + with moderate prognosis

Clinical picture:

1. Arthritis (is Swelling or effusion OR 2 of the following signs: hotness, tenderness,decreased range of movement) , no redness.
2. Gelling (stiffness after periods of rest, such as long car rides), morning joint stiffness, and pain.)
3. In Young Child :Intermittent limp and deterioration of mood or behaviour
4. Presentation initially can be indolent , especially in young children minimal or no swelling – Ix CBC – CRP normal
5. If systemic Features or present sepsis and malignancy must always be considered.

Complications:

1. **Chronic Anterior Uveitis**:Common – Asymptomatic – can lead to loss of vision , Regular ophthalmology screening with slit lamp especially in oligo arthritis and ANA +ve / HLA B27 .
2. **Joint deformities, limb length discrepancy, flexion contractures of joints preventable by early D and Rx**
3. **Growth failure due to JIA or CS**
4. **Constitutional symptoms Anemia – Delayed Puberty**
5. **Osteoporosis :Ca + Vitamin D is important**
6. **Macrophage activation syndrome**:The most serious complication of systemic JIA is macro– phage activation syndrome (MAS), which occurs in 4%–13%
7. **Amyloidosis :Cause proteinuria ,ARF**
8. **Psychological complications & poor school performance**

Treatment: MDT .

1. NSAID + Analgesics: Relieve Symptoms only & do not modify disease
2. Joint injection: Effective – 1st line for oligoarticular .
3. Methotrexate:
    - Early use decrease joint damage
    - Effective in polyarthritis 70%
    - Oral or Injection
    - Nausea & vomiting are common

4. Systemic C S
    - Ovoid if possible
    - IV CS is given in severe polyarticular , So JIA or macrophage activation syndrome
5. Cytokines Modulator – Immunotherapy (infliximab –Anakinra)
6. Orthopaedic –physiotherapy –Ophthalmology– dentistry School – Social worker

Prognosis:

- Early aggressive treatment improves outcomes and prevents complications such as joint damage or contracture, limb length discrepancy, and blindness from uveitis.
- In spite of immunomodulatory medications, more than half have active disease into adulthood

Kawasaki Disease "KD"

"Mucocutaneous lymph node syndrome"

- It is acute febrile vasculitis involving medium sized blood vessels with special predilection of the coronaries
- It's the commonest cause of acquired heart disease in UK
- Ch⁻ by [Disease of childhood (80% <5 year) – Clinical diagnosis
- Etiology: Unknown [Immune Mediated] may be infections genetic
- Epidemiology: Sex (more in ♂) – Age (6 month– 4 years) peak end of first year – more in Asian Japanese

Clinical picture:–

Diagnostic Criteria:

1. **Fever for at least 5 days** (>5days)
2. The presence **4 of the five following features"**
   - 1.**Bilateral bulbar conjunctival injection** (red eyes) without exudate(discharge).
   - 2. **Erythema and cracking of lips, strawberry tongue and/or erythema of oral and pharyngeal mucosa**
   - 3. **Cervical lymphadenopathy** (≥1.5 cm diameter), usually unilateral.
   - 4.**Rash:maculopapular,** diffuse erythroderma, or erythema multiforme–like.
   - 5.**Erythema and oedema** of the hands and feet in acute phase and/or **desquamation"peeling** " around the nails in subacute phase.



Other Clinical Manifestations:

- CVS
  - Myocarditis and HF (≥ 4wk)
  - Pericarditis ± effusion
  - Coronary arteries Aneurysm in 20 – 30% (more in infants)
  - MI and infraction
  - Rayounds phenomena
- CNS: Irritability – aseptic meningitis
- GIT :Hepatitis – Abdomen pain and diarrhea
- Joints: Arthritis / Arthralgia

Clinical Phases:

	Duration	Main Features
Acute Febrile Phase	1st 1–2 weeks	Fever – myocarditis and erythema
Subacute Phase	Till the 4th week	Desquamation – Coronary Aneurysm
Convalescent Phase	Till the 6th – 8th week	No Clinical signs only increase ESR and CRP

Investigation: "No specific lab test used for diagnosis of KD"

1. CBC: WBC high – PLt high > 1 week
2. ESR and CRP high
3. LFT: high ALT– AST
4. Urinalysis: sterile pyuria
5. Echo [at diagnosis ,2–4week ,6–8 weeks,6–12 weeks]
6. ECG – Chest X Ray

Treatment:

1. IVIG "DOC" (2g/kg over 10 hour) single dose (decrease Risk of coronary aneurysm in 1st 10 days)
2. Aspirin:High dose 80–100mg/kg/d for 14 days the 3–5 mg/kg/d once daily for 6–8 weeks after the onset of disease [ESR :N – Temp : N]
3. Follow up in cardiology OPD is very important
4. Refractory KD: Steroid – infliximab – cyclosporine.

Prognosis: "MR 0.5–1 %"

- Patient with No coronary abnormalities: complete recovery
- Patient with coronary abnormalities: according to severity
  - Resolution ≃ 50% within 1–2 years and
  - If giant unlikely to resolve (give warfarin or low dose aspirin)
- Poor Prognosis features are ……
- Covid 19 ??! Relation .
- Incomplete KD: occur usually in infants < 6m + prolonged fever ,but not all the cardinal features are present. ??

"Systemic Lupus Erythematosus SLE"

- Is chronic multisystem autoimmune disease affecting many organs (skin, joints, kidney, CNC, Lungs..)
- **Etiology**: Unknown [Genetic – Hormonal – Environmental factors]
- **Epidemiology**:
- **Age**: >5 years
- **Sex**: ♀>♂
- Onset Acute gradual
- Course remission and relapse

- **Diagnostic criteria** (4/11):

1. Malar rash (sparing nasolabial folds)
2. Discoid Lupus
3. Photosensitivity
4. Oral ulcer (painless)
5. Arthritis (non erosive ≥2joints)
6. Serositis :pleurisy– pericarditis
7. Nephritis (Proteinuria + cast: RBC – granular)
8. Haematological (Haemolytic Anemia – low WBC – lymphopenia – low plt)
9. Neurological – seizure – Psychosis
10. Positive ANA (most sensitive)
11. Positive Immunoserology [Anti dsDNA Ab (most specific) ,Anti Smith Ab and Anti phospholipid Ab]

- **Other Criteria:**
  1. **Clinical**
     - Constitutional/Weight loss –Lymphadenopathy – malaise
     - MSK: Arthralgia – Myalgia
     - Skin: Alopecia – Raynaud's phenomenon
     - Pulmonary: Pneumonia – Hge – Embolism – HTN
     - Cardiac: Myocarditis
     - CNS: Headache – optic neuritis – CVA.
  2. **Lab**
     - High ESR – low C3 – low C4

- **Treatment**
  1. Cornerstone Rx [Corticosteroid– Cyclophosphamide – Azathioprine] Rituximab – IVIG
  2. Symptomatic Rx: Avoid sun exposure – sunscreen, NSAID for Arthritis Topical steroid
  3. Rx of complication as dialysis in lupus nephritis, Anti HTN drugs

- Major Causes Death: Renal – CNS – Infection – Pulmonary Hge.

Neonatal Lupus Erythematosus

- Definition: Disease occur in neonate born to mother with SLE or Sjögren Syndrome SS.
- Incidence: 100% of SS mothers, < 3% of SLE mothers
- Pathogenesis:Affected mother (even asymptomatic) → Auto–antibodies → Pass placenta: Anti Ro antibodies– Anti La antibodies
- Clinical picture: Male = Female
    - Skin :Annular erythematous macules
    - CVS: Congenital Heart Block (CHB), Prenatal bradycardia → mistaken as fetal distress
- Monitoring: Fetal ECG
- Postnatal pacemaker for all affected infants ± Heart transplantation



Genetic & Skeletal Conditions

1. Osteogenesis imperfecta (Brittle bone disease):

- It is a group% of disorders of collagen 1 metabolism causing bone fragility with bowing and frequent fractures
- AD
- C/P Triad of
    - Blue sclera
    - fragile bone with easy fracture
    - hearing loss (otosclerosis)

+

- Many types: Type I is the most common type & Type II: Is the most severe type (fatal) and multiple ↑ before birth or stillborn
- Complications: recurrent infection, Hydrocephalus, compression.
- Rx: Bisphosphonate (decrease bone resorption), Rehabilitation. Manage fractures & deformities

2. Achondroplasia:

- AD (50% new mutation) (♂=♀)
- Gene mutation: Fibroblast Growth Factor Receptor 3 (FGFR3)
- C/P
    1. Short stature (short limbs)
    2. Large Head (Macrocephaly)–Hydrocephalus.
    3. Frontal bossing, depressed nasal bridge ,recurrent OM.
    4. Hands short and broad ,Trident fingers,,,Brachydactyly
    5. Lumbar lordosis, scoliosis











3–Osteopetrosis (Marble Bone Disease العظم الرخامي):

- Rare disorder
- AR (severe type)
- Bone dense but brittle
- C/P :FTT –recurrent infection–HSM – Hypo ca – Anemia– thrombocytopenia
- Prognosis :Poor
- Bone marrow transplantation can be curative



Summary

Diagnostic Clues regarding musculoskeletal disorders

Typical Symptom Combinations	Pivotal Clinical Features	Possible Diagnoses
Nocturnal Waking with leg pain	Normal Child Anaemia Bruising, irritability, infections	Growing pains Osteoid Osteoma Leukaemia, Lymphoma, Neuroblastoma (Young child)
Clunk on hip movement on neonatal and early infant screening, limp in an older infant	Older infant asymmetrical upper leg skin folds, limited hip abduction	Developmental dysplasia of the hip (DDH)
Febrile, toxic – looking infant, irritability with nappy changing	Restricted joint range(especially hip) or limb movement	Septic Arthritis Osteomyelitis
Sudden limp in an otherwise well young child	Unilateral restricted hip movement	Transient synovitis of the hip perthes disease
Fever, erythematous rash, red eyes, irritability In infant or young child	Erythema/Oedema of hands and feet, oral mucositis, cervical lymphadenopathy	Kawasaki disease
Irritability, fever, reluctance to move in an infant or young child	Stiff back 'tripod' slitting	Discitis Vertebral Osteomyelitis
Joint pain, stiffness, and restriction Loss of joint function	Persistent joint swelling loss of joint range	Juvenile Idiopathic arthritis
Hip pain in an obese adolescent boy	Unilateral hip restriction	Slipped capital femoral epiphysis
Lethargy, unwilling to do physical activities, irritability, rash	Eyelid erythema Proximal muscle weakness	Juvenile dermatomyositis
Constitutional symptoms, lethargy, arthralgia in an adolescent female	Multisystem abnormalities, haematuria, facial erythema	Systemic lupus erythematosus

Inborn Errors of metabolism "IEM"

- **IEM:** Are a group of genetic disorders of metabolism most of which are due to defects of single genes that code for enzyme
- IEM are individually rare ,but relatively common as a group
- [A + B Enzyme C]
- Most of IEM are Autosomal recessive "**AR**" Except: [**Hunter's, Fabry's disease, Lesch– Nyhan disease , Ornithine transcarbamylase deficiency (OTD), Adrenoleukodystrophy are XLR**]

Clinical Picture:

1. Before birth : Positive F/H or Previous Unexplained Death in family ,H/O consanguinity.
2. At Birth : Usually Normal
3. In Neonatal period and Children
    - Vomiting – poor feeding ,FTT ,diarrhea
    - Neonatal Convulsions
    - Encephalopathy (rapidly progressive coma)
    - Hepatomegaly (organomegaly)
    - Prolonged neonatal Jaundice
    - Poor neonatal activity
    - Characteristic (peculiar) Urine odour
    - Developmental delay ,dysmorphic features .
    - Hypoglycemia – metabolic acidosis
    - unexplained respiratory alkalosis
    - High ammonia
    - Poor response to conventional therapy
    - cardiac failure or cardiomyopathy
    - sudden unexplained death .

Classification:

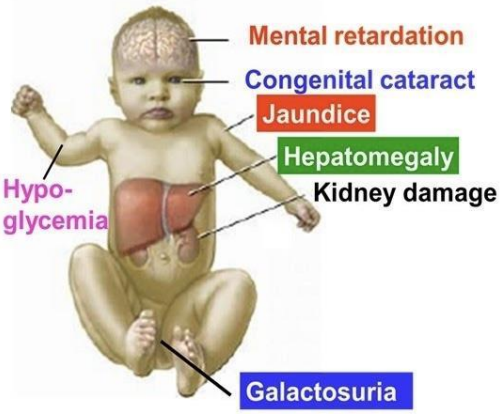
1. Disorders of CHO metabolism:
    - Galactosemia
    - Glycogen Storage disease
    - Muco polys accari dosise
2. Organic Acidemias:
    - Maple Syrup Urine disease
    - Isovaleric acidemia
    - Propionic Acidemia
3. Disorders of Amino acid metabolism:
    - Phenylketonuria
    - Tyrosinemia

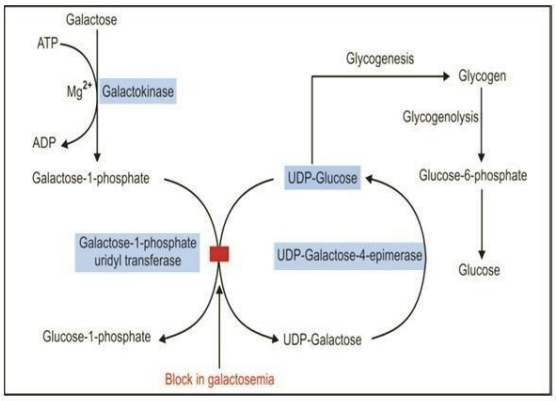
- Homocystinuria
- Alkaptonuria

4. Urea cycle disorders:
    - Ornithine transcarbamylase deficiency (OTD)
    - Argininemia
    - Other rare types
5. Disorders of Fatty acid Oxidations mitochondrial metabolism
    - Glutaric acidemia
    - Medium chain acyl – COA dehydrogenase decrease
6. Lysosomal Storage disease:
    - Gaucher's disease
    - Niemann – pick disease
    - Fabry disease
    - Tay –sachs disease
    - Wolman disease
    - Metachromatic Leukodystrophy

"Galactosemia"

- Most common IEH CHO disorder
- Its rare AR disorders (1:23,000–40,000)
- Its due to decrease of galactose 1– phosphate uridyl transferase (GALT)
- Normally: Galactose Galactokinase Galactose 1 phosphate gla – 1 – P Uridyle Transferase Glucose
- Lead to accumulation of galactose 1 p in the Liver, brain, eyes and kidney





≃

C/P:

1. Poor Feeding – vomiting – diarrhea – FTT
2. Jaundice (cholestatic) – Hypoglycemia – lethargy
3. Cataract (high glacticol)
4. Hepatomegaly and liver failure
5. Severe neonatal sepsis by(E Coli)Death
6. Intellectual Disability
7. Fanconi syndrome

Diagnosis:

1. Clinically
2. Newborn screening
3. Urine for reducing substances +ve
4. Definitive Diagnosis (GTP in RBCs)

Rx: Galactose free diet

Complication:

1. E. Coli sepsis
2. Developmental Delay
3. Learning disability [language and speech] deficient even with good Rx
4. Ovarian Failure

"Glycogen Storage Disease"

- Its group of disease due to specific enzyme defects preventing metabolism of glucose from glycogen and this leads to abnormal storage in liver and or muscle.
- Types: "All are AR disorders"

Table some of the glycogen storage disorders

Type	Enzyme Defect	Onset	Liver	Muscle	Comments
Type I(Von Gierke's)	Glucose 6 – Phosphatase	Infant	+++	–	Increase TG Doll like face Enlarged liver and kidneys growth failure hypoglycaemia good prognosis
Type II (pomps)	Lysosomal αGlucosidase	Infant	++	+++	Hypotonia and cardiomegaly at several months death from heart failure
Type III (Cori)	Amyio 1,6 glucosidase	Infant	–+	+	Milder features of type I but muscles may be affected good prognosis
Type IV	Anderson disease				
Type V (McArdle) most common type	Phosphorylase	Child		++	Temporary weakness and cramps muscles after exercise myoglobinuria in later site

Investigations:

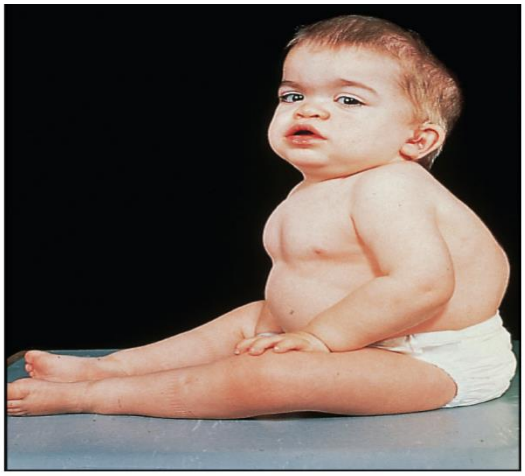
- RBS
- Liver biopsy
- MS biopsy
- Genetic study
- LFT – ECG

Rx:

1. Maintain blood glucose by frequent feeds
2. Corn starch (slowly releasing)
3. Enzyme replacement therapy (Pompe)
4. Liver transplantation

Mucopolysaccharidosis (MPS)

- Group **lysosomal** disorders. There is defective breakdown of **glycosaminoglycans (GAGs)** & accumulation in organs
- They are **progressive multisystem** disorders which may affect the CNS, ocular, cardiac, and skeletal systems.
- **All MPS are AR except Hunter's syndrome is XLR**
- MPs have normal development initially and abnormalities seen in infancy or childhood



C/P:

1. **Face** :coarse facies – large tongue – large head– thick eyebrows
2. **Eyes** :Corneal clouding [No in hunter's] (skin nodule on scapula)
3. **Heart**:Valvular lesions [No in sanfilippo]
4. **Skeletal**:Broad ribs – thoracic kyphosis – lordosis –claw hands – thickened skull
5. **Brain**: Developmental delay and MR [No in morquio and Maroteaux]
6. **Other** :Hepatosplenomegaly – umbilical and inguinal hernia –Conductive deafness

Types:

Table 27.8 Types of mucopolysaccharidoses

Type	Inheritance	Cornea	Heart	Brain	Skeletal
MPS I (Hurler)	AR	+++	++	+++	++
MPS II (Hunter)	X-linked	–	+	++	+
MPS III (Sanfilippo)	AR	±	–	+	+
MPS IV (Morquio)	AR	+	+	–	+++
MPS VI (Maroteaux–Lamy)	AR	+++	++	–	++

AR, autosomal recessive.

Investigations

- Urinary excretion level of the major storage substances, the glycosaminoglycans (GAGs).
- Lysosomal enzyme testing or genetic testing.

Rx:

1. Enzyme replacement therapy
2. BMT (haemopoietic stem cell transplantation)
3. Orthopedic Surgery

Hyperammonemia

Ammonia is a highly toxic chemical derived from bodily nitrogen. It is detoxified to urea by the urea cycle, which principally occurs in the liver. An ammonia level should be measured when there is:

- Unexplained encephalopathy
- Respiratory alkalosis because it is a respiratory stimulant
- Recurrent vomiting
- Unexplained severe illness in a baby or child
- Unexplained seizures as it causes cerebral oedema.

Ammonia is elevated in :**urea cycle defect –organic acidemia – severe illness, liver disease, by certain medications, and transiently in the newborn.**

Treatment of Acute Hyperammonemia (should be rapid & aggressive)

1. Adequate Calories:IV fluid: electrolytes & Glucose 10%
2. Protein restriction
3. Enhancement of ammonia excretion
   - Na benzoate: Removes NH in the form of hippuric acid
   - Phenylacetate: Removes NH, in the form of phenylacety|glutamine
   - Arginine: Except in arginase deficiency
4. Dialysis :If there is no significant decrease of NH3, Hemodialysis is preferred
5. Oral neomycin :to decrease intestinal production of NH3
6. Oral Lactulose:to decrease Intestinal absorption of NH3.

"Organic Acidemias"

- Its rare AR disorders which disrupts or prevents the normal metabolism of the branched chain amino acids (Leucine – Isoleucine – Valine)
- **Common types:**
  1. **Isovaleric Acidemia** (isovaleryl–CoA dehydrogenase deficiency)
  2. **Methylmalonic acidemia** (methylmalonyl–CoA mutate deficiency)
  3. **Propionic acidemia**(propionyl–CoA carboxylase :biotin cofactor deficiency
  4. **Maple syrup urine disease (MSUD)** : branches–chain dehydrogenase deficiency

C/P:

Healthy newborn

- Rapidly ill – poor feeding – vomiting – hypotonia
- Tachypnoea – Convulsions –opisthotonus
- Encephalopathy – coma – death
- Severe metabolic acidosis– Hypoglycaemia – high Ammonia
- Specific odour
  - Isovaleric Acidemia : Sweaty feet
  - MSUD : Maple syrup

Ix:

- RBS: Low
- AG :Metabolic acidosis (high anion gap)
- Ammonia : high
- CBC: pancytopenia
- Plasma Leucine – isoleucine – valine :high in MSUD
- Enzymes assay & gene analysis
- Neonatal screening test :+ve

Rx:

- ABC–call for help
- IVF fluid .
- IV dextrose and NaHCO3 for acute attack
- Treatment of hyperammonemia: Na benzoate –or phenylacetate– or dialysis in severe cases .
- Restriction of protein diet –adequate calories
- Carnitine supplementation

Urea cycle defect

The function of urea cycle is to get rid of ammonia

Five enzymes are involved in synthesis of urea:

- Carbamoyl phosphate synthetase (CPS)
- Ornithine transcarbamylase (OTC)*
- Citrullinemia: Argininosuccinate synthetase (AS):
- Argininosuccinate lyase (AL): Argininosuccinic aciduria
- Arginase: Hyperargininemia

All are AR except Ornithine transcarbamylase (OTC)* mcq

Epidemiology 1: 30.000 live births

Clinical Picture:–

Neonatal period	Infants & older children
• Normal at birth • Poor feeding, vomiting, tachypnea • Lethargy, coma, convulsions, high ICP • Hepatomegaly	• Vomiting • CNS: Ataxia, confusion, irritability (alternating with lethargy & coma)

Investigations

- High ammonia level : Normal values: < 35 (children), < 100 (full term), < 150 umol/L (Preterm).
- LFT : ALT, AST high
- ABG: Respiratory alkalosis.
- High Metabolites substances according to type .

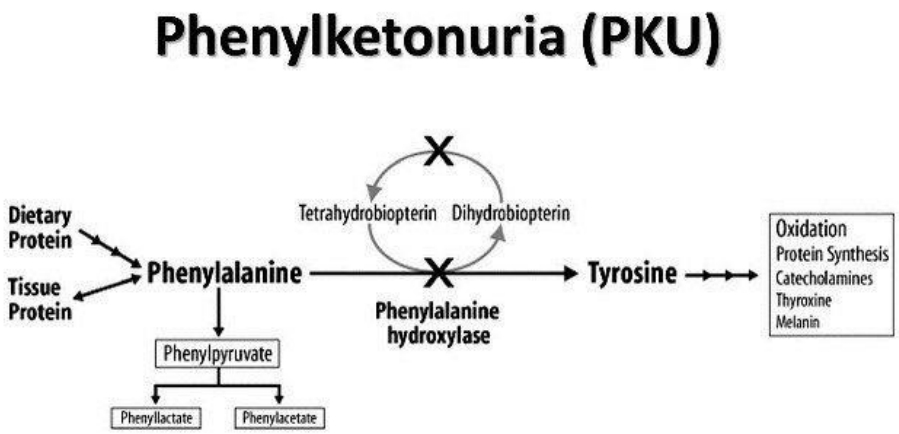
Treatment : as mentioned in treatment of acute hyperammonemia (box) .

- Long–Term treatment of Hyperammonemia(Adequate Calories, protein restriction)
- Enhancement of ammonia excretion(Na benzoate,Phenylacetate,Arginine –Citrulline: In OTC deficiency)
- Carnitine supplementation
- Avoid triggering factors: Infections, fasting

Disorders of AA metabolism

1. Phenylketonuria "PKU"

- "Most common" IEM of AA 1:10,000
- AR disorders and phenylalanine, Tyrosine + Tryptophan
  ◂Hydroxylase



C/P: usually normal at birth –

- at age 6 – 12 months:
  1. Fair skin and hair(light complexion) – Eczema – Light sensitivity
  2. Hair loss
  3. Increase Risk of pyogenic infection

Other C/P if treated PKU:

1. Intellectual disability ,microcephaly
2. Epilepsy (50%) and Extrapyramidal manifestation
3. Musty or mousy odor
4. Eye Abnormalities (blue eyes)

Diagnosis:

1. Neonatal Screening test (TMS)
2. Elevated phenylalanine level >20 ng/dl

Rx:

1. Phenylalanine restricted diet for life (target level 2–6ng /dl)
2. Essential amino acids – vitamin – minerals supplementations

Maternal PKU : Microcephaly –MR –CHD .(phenylalanine is teratogenic)

2. Homocystinuria

- Rare AR (1:200,000)
- Due to deficiency of Cystathionine synthase enzyme

C/P:

1. Marfanoid features tall stature
2. Developmental delay – MR
3. Limited joint mobility
4. Lens dislocation (Downward and medially)
5. Increase Risk of thromboembolism

Diagnosis: – Homocystinuria serum methyl malic acid (specific)

Rx:

1. Large dose of pyridoxine – folic acid
2. Low methionine diet

	Marfan syndrome	Homocystinuria
Etiology	Defect in collagen fibers	Cystathionine synthase I
Inheritance	AD	AR
Mentality	Normal	MR
Musculoskeletal	Arachnodactyly (2 Tests), Pectus excavatum, High arched palate, Kyphoscoliosis +Hernias & Pneumothorax	The same (Marfanoid) Arachnodactyly Pectus excavatum, High arched palate, Kyphoscoliosis
Bone density	Normal	Osteoporosis
Joints	Lax	Stiff
Cardiovascular	AR, MR, Aortic dissection	No or mild (AR, MR)
Ocular	Lens dislocation (Upwards)	Dislocation (Downwards)
Vascular thrombosis	No risk of thrombosis	Increase risk of thrombosis
Investigations	No Homocysteine in urine	High Homocysteine in urine
Treatment	Supportive	Low methionine diet B6 & folic acid

3. Tyrosinemia:

- **Tyrosinemia (type I): infantile type** مهمة
  - Is a rare **autosomal recessive** disorder
  - Low **Fumarylacetoacetate hydrolase enzyme** → high Succinylacetone → Organ damage (liver failure – renal tubule (Fanconi syndrome & peripheral nerves)
  - If untreated: **fatal**
  - C/P start in the first few months of life (2–6 months)

Liver	kidneys	peripheral nerves
• Fever, irritability, vomiting, hepatomegaly, jaundice, hypoglycemia, coagulopathy. • Boiled cabbage oder (high Methionine) • Liver cell failure, cirrhosis, hepatocellular carcinoma (> 2 yrs)	• Proximal RTA • Vitamin D–resistant rickets	• Pain, hypertonia, paralysis

Investigation

- High Blood & urine succinylacetone
- High alpha Fetoprotein
- Investigation of liver, hematologic & renal affection: ALT, AST, bilirubin, glucose, CBC
- Neonatal screening (succinylacetone)
- Prenatal diagnosis is available (AF succinylacetone or CVS)

Treatment

- Phenylalanine & tyrosine restricted diet
- Nitisinone (NTBC)
- Liver transplantation

Tyrosinemia (type II): oculo–cutaneous

- AR – defect in hepatic tyrosine aminotransferase gene
- C/P:

1. Eye (Photophobia), skin lesion (palmoplantar keratosis)
2. Neurological and developmental delay– Microcephaly

- Rx: Low tyrosine and phenylalanine dies

Tyrosinemia (type III) :

- Hydroxyphenylpyruvate dioxygenase deficiency
- Seizures, MR, ataxia–Self mutilation
- Rx:diet +vit

"Lysosomal storage disease"
Sphingolipidoses

- Are class of **lipid storage disorders** relating to sphingolipid metabolism.

Disease	Deficient Enzyme	Inheritance	Symptoms	Rx and prognosis
Gaucher Disease	B glucosidase enzyme Age 2 –18 years	AR Diagnosis: BMA(Gaucher cells) level of enzyme	splenomegaly Bone pain, Pancytopenia, Bone deformity , severe Anemia bleeding – Normal IQ	Enzyme replacement therapy MBT Gene therapy
Niemann pick disease	Sphingomyelinase deficiency	AR Types : Infantile Juvenile Adult	MR, Spasticity, Seizures, Hepatosplenomegaly , low plt, Ataxia	Limited Rx death by the age 7 yr to adulthood
Fabry disease	Alpha – galactosidase A	X L R	Acroparesthesia, hypohidrosis, Angiokeratomos, Corneal opacity Normal IQ	Enzyme replacement therapy.
Tay–sachs disease (GM2 Gangliosidosis)	Hexosaminidase A (No visceral involvement)	AR	marked startle to noise or light, Neurodegeneration, early death, cherry red spots	No Rx Die by 4 years

"Poisoning"

Types:

1. Acute poisoning :Accidental Poisoning, Suicidal poisoning and Iatrogenic poisoning
2. Chronic Poisoning

Background:

- Child <6 years have the greatest risk
- Mostly involve single substance
- Mostly occur in the home ≃90%.
- Ingestion is the most common route of poisoning Other route – skin – inhalation
- Repeated poisoning may be sign of family problem
- Most common agents responsible for poisoning [Paracetamol – Aspirin – TCA – Iron] Organ phosphorus – Hydrocarbon
  -

1. Paracetamol ingestion [Acetaminophen]

- It is the most widely used analgesic and antipyretic
- Toxicity results from the formation of highly reactive intermediate metabolite(N–acetyl–p–benzoquinoneimine) (NAPQI)
- Toxic Dose [>200 mg/kg] in children <12 years .

C/P:

- Stage I :1st 24 hr: Asymptomatic – Non–specific (N – V–pallor – diaphoresis)
- Stage II :24– 48 hr: resolution of the symptoms
- Stage III :> 48 – 96 hr: LFT increased – high HR – low Bp and Liver failure – Encephalopathy
- Stage IV :4 days –3wks: complete recovery or death

Management:

Ix:

- ABC –call for help
- Prevent absorption: gastric lavage – activated charcoal
- Laboratory : (glucose–electrolytes –ABG –RFT –LFT –coagulation profile –Paracetamol serum level > 4 hr (Rumack–Matthew nomogram).
- Anti–dote (N – acetylcysteine IV or oral in 1st 24 hr after ingestion (1st 8 hours is preferred & more effective) .

2. Aspirin Ingestion (salicylic acid)

- Toxic Dose [>150 mg/kg]
- Salicylate affect most organ systems by
    - Uncoupling oxidative phosphorylation
    - Inhibiting krebs cycle enzyme
    - Inhibiting AA synthesis
    - Decrease platelets adhesiveness
    - Increased pulmonary capillary permeability.

C/P:

- Nausea – vomiting – diaphoresis –Tinnitus
- Hyperventilation(directly stimulate respiratory center),leads to **respiratory alkalosis**
- **Metabolic acidosis**(because accumulation of lactic acid)
- CNS: Agitation – confusion–seizure– coma
- Hyperthermia

Investigation:

- ABG – U/E – RBS …?!
- LFT – Coagulation study
- Serial serum salicylate level / 2 hr

Rx : ABC – Decontamination – Na HCO3(urinary alkalinization)– if severe dialysis

3. Organophosphorus Poisoning:

- Usually result of accidental exposure to insecticides in and around the home or farm .
- It's very dangerous and may lead to death
- It may absorbed through skin, lung – GIT
- It bind to cholinesterase enzyme preventing the degradation of ACH resulting in its accumulation at nerve synapse

Clinical Picture:

1. **Muscarinic**
    - ✓ Vomiting – Urination – diarrhea
    - ✓ Lacrimation – sweating–excessive salivation
    - ✓ Miosis – Bronchospasm
    - ✓ Bradycardia – low Bp

DUMBBELLS : D– Diarrhea , U – Urinative , M– Moises , B– Bradycardia , B – Bronchospasm ,E– Emesis,L –Lacrimation,L –Low Bp , S–Salivation

2. **Nicotinic:**
    - ✓ MS weakness
    - ✓ Fasciculation
    - ✓ Tremor
    - ✓ Hypertension–tachycardia

3. CNS :Confusion – delirium – seizure – coma

Management:

1. ABC
2. Decontamination:gastric lavage , Wash all skin with water & soap and remove cloths
3. Antidote :Atropine and pralidoxime

4. Hydrocarbon Ingestion: (Kerosine – gasoline)

- ✓ Aspiration of small amount may lead to serious and life throating toxicity
- ✓ Pneumonitis due to aspiration is the most important effect [Cough – Increase RR – RD Signs – fever]
- ✓ May lead also to CNS inhibition – hepatic and renal toxicity
- ✓ Investigation :C X Ray

Rx:

- ABC
- Observation and supportive care
- Respiratory support
- Steroid and antibiotics (……role ?!)
- GL –emesis –charcoal are CI .

5. Iron ingestion:

- ✓ Is common cause of childhood poisoning death
- ✓ The potential severity of exposure is based on the amount of elemental iron ingested
  - Local effect on GIT(corrosive)
  - Vasodilation and increased permeability leading to hypotension
  - Metabolic Acidosis
  - Hepatic necrosis: abnormal LFT
  - Drowsiness, coma: .Hemodynamic instability,direct toxic effect of iron in CNS
- ✓ Toxic dose (>60 mg/kg)

C/P:

1. Stage I:Gastro intestinal stage < 6 hr – N– V, Abdomen pain, hematemesis and diarrhea
2. Stage II :Stability stage (6 – 24 hr)
3. Stage III :Systemic toxicity (within 48hr)
   – Cardiovascular collapse
   – Severe metabolic acidosis
   – Coma ,Hepatotoxicity and liver failure (2 –3 days)
4. Stage IV: 2–4 weeks ,gastric scarring ,pyloric stenosis .

Investigation

Abdominal X Ray(iron is radio opaque) – ABG , Serum iron level and U/E – RBS – LFT – coagulation profile

Treatment: ABC

- GL & charcoal are ……?
- Desfroxamine IV infusion in moderate to severe cases .

6. Tricyclic Antidepressant (TCA)

- Block reuptake of norepinephrine, serotonin, dopamine in CNS, peripheral nervous system
- **Toxic dose** is 5–20 mg/kg

C/p: affect mainly CNS, CVS

- CNS: Drowsiness, Lethargy ,Coma,Seizure
- CVS: Tachycardia, Hypertension , Arrhythmia(VT,QT prolongation, ST segment depression, complete H.block)

Investigation: – ECG monitoring – Blood level(diagnosis)

Management: – ABC

- General poisoning measures
- Treatment of dysrhythmia
- Asymptomatic child should be kept under observation at least 6 hrs. with ECG monitoring

Common Antidote for Poisoning: MCQ

Poison	Antidote
Acetaminophen	N – Acetylcysteine (mucomyst)
Anticholinergics	Physostigmine
Benzodiazepines	Flumazenil
β –blockers	Glucagon
Calcium channel blockers	Insulin and calcium salts
Carbon Monoxide	Oxygen
Cyanide	Nitrates
Digitalis	Digoxin – specific fragments antigen binding(Fab) Antibodies
Ethylene Glycol and methanol	Fomepizole
Iron	Deferoxamine

Isoniazid (INH)	Pyridoxine
Lead and other heavy metals, eg mercury and arsenic	BAL (Dimercaprol)
Methemoglobinemia	Methylene blue
Opioids	Naloxone
Organophosphates	Atropine and pralidoxime
Salicylates	Sodium bicarbonate
Sulfonylureas	Octreotide
Tricyclic antidepressants	Sodium bicarbonate
Warfarin	Vitamin K
Heparin	Protamine Sulphate

Drowning

Drowning is a major cause in head injuries and death

- Initial Peak – Toddler age group
- Second peak – Male Adolescents
- Children 1–4 years of age:
  - – Likely down in swimming pools where they have been unsupervised temporarily (Usually for <5min)
  - – Typical incidents involve a toddler left unattended temporarily or under the supervision of an older sibling
- Adolescent and young adult age groups (ages 15 – 24 years) – Most incidents occur in natural water
- Approximately 90% of drowning occur within 10 yards of safely
- Parent should be within an arm's length of a swimming child (anticipatory guidance

Mechanism of injury:

- Initial swallowing of water
- Laryngospasm
- Loss of consciousness
- Hypoxia
- Loss of circulation
- Ischemia
- CNS injury (The most common cause of death)
- Acute respiratory distress syndrome (ARDS) may develop
- Salt water drowning classically associated with
  - ✓ Hypernatremia
  - ✓ Hemoconcentration
  - ✓ Fluid shifts and electrolyte Not clear clinically

- Fresh water drowning classically associated with
  - ✓ Hyponatremia and hemodilution
  - ✓ Hyperkalaemia
  - ✓ Haemoglobinuria and renal tubular damages

Management of drawing and near drawing

- Cardiopulmonary resuscitation(CPR) at the scene
- Admit regardless of clinical status
- All children with submersion should be monitored in the hospital for 6–8 h
- If no symptoms develop can be discharged safely
- 100% oxygen with bag and mask immediately
- Nasogastric tube for gastric decompression cervical spine immobilization if suspected cervical injuries
- Positive end expiratory pressure (PEEP) and positive pressure ventilations in case of respiratory arrest
- Continuous cardiac monitoring
- Bolus of normal saline or Ringer's lactate
- Vasopressors
- Defibrillation if indicated

"Reye syndrome"

- Reye syndrome is characterized by acute non– inflammatory encephalopathy and fatty degenerative liver failure
- Typically occurs viral illness (VZV – influenza – AGE) and is associated with use of aspirin during the illness

Epidemiology:

1. Age: 5–15 years
2. Sex : ♀=♂

Pathophysiology: Unclear (Mitochondrial injury)

C/P:

1. H/O viral URTI 3 weeks
2. Abrupt onset of vomiting – lethargy , Hyperventilation – irritability – delirium seizures – coma

Examination

1. Vomiting with or without dehydration
2. Hepatomegaly
3. Jaundice ±
4. Seizure – Coma – Paralysis

Investigations:

1. LFT :increase ALT increase AST
2. Ammonia level::increase 1.5 times normal
3. PT and PTT :Prolonged in 50%
4. RBS :usually normal but may low
5. RFT :high
6. ABG:Metabolic acidosis

Rx:

1. PICU Rx supportive care
2. Monitoring continuous and correct RBS and acidosis

"SIDS" sudden infant death syndrome

Definition:

- Is sudden death of an infant younger than 1 year for which no adequate cause is found after a thorough post–mortem examination.

Factors associated with SIDS: mcq

I. **The infant**
- ✓ Age 1–6 months (peak 12 weeks) (2–4 months)
- ✓ LBW – preterm (60% term e⁻ normal birth weight)
- ✓ Sex (increase boys 60%)
- ✓ Multiple births
- ✓ H/O SIDS in sibling increase Risk

II. **The Parents**
- ✓ Low income – poor or overcrowded housing
- ✓ Maternal age (young mother < 20 years risk) j
- ✓ High maternal parity
- ✓ Single unsupported mother (2x)
- ✓ Maternal smoking during pregnancy
- ✓ Parental Smoking after baby's births

III. **The environment**
- ✓ The infant sleeps lying prone
- ✓ The infant is overheated from high room temperature (more in winter)

Management of the sudden unexpected death of an infant:

1. Resuscitation :take to the ER initiate resuscitation unless in appropriate
2. Care of parents:Should be cared by specific member of staff and History should be obtained
3. Baby pronounced dead :Detailed clinical Exa by consult

- – Remove ETT and IDL but retain IV livers
  – Retain child's clothes and nappy for police
  – Ix

1. Nasopharyngeal aspirate
2. B1 C and S
3. B1 for toxicology – metabolic screen – chromosome
4. Urine for bio – tox edogy
5. Lp for virology – culture

– Paediatrician – police 1ry care team informed

4. Breaking the news to the parents:
   – Performed by paediatrician explain that the police will be involved
5. Parents offered to see and hold their baby
6. Home visit within 24 hrs

"Immunology"

- **Component of immune system:**
  - **Innate Immunity** :Produce specific and nonspecific :
    a. Phagocyte
    b. Complement
    c. Soluble mediation
  - **Specific immunity** :Produce specific response against an antigen:
    1. 1ry Lymphoid tissue [B cells (BM)– T cells (Thymus)]
    2. 2ry Lymphoid tissue [Lymph node – spleen]

- **T Lymphocyte:**
  1. T helper cells CD4
  2. T Cytotoxic cells CD8
- **B Lymphocyte :Produce Immunoglobulin (Ab)**

1. **IgG** :

   - Major serum Ig 75%
   - Can cross the placenta
   - Ab of 2ry response
   - 4 subclasses IgG1, IgG2, IgG3– IgG4
   - T1/2 = 21 Days

2. **IgM** :

   - Major Ab of 1ry response (Acute infection)
   - Dose not cross the placenta

3. **IgA**:Secretory Ig ,Protect mucosal surface
4. **IgD**:present on B cell membranes, precise function unknown
5. **IgE**:Involved in type I hypersensitivity and against helminth parasites

"Immune Deficiency"

1. Primary (uncommon) – a genetically determined defect in the immune system.
2. Secondary (common)– Causes
   - Malnutrition
   - Malignancy Leukemia – Lymphoma BM Aplasia
   - Infection:Viral – HIV, Measles, CMV , EBV and Bacterial infection
   - Drugs: Immunosuppressive Therapy – chemotherapy – steroid (High dose long peruses)
   - Disease: N S – D M, Zinc decrease – Uraemia – Burns, protein losing enteropathy–SLE.
   - Others : Splenectomy and BMT

- **Immune deficiency:**
  Should be considered A child who presents with [Sever. Prolonged, Unusual or Recurrent infections (SPUR)]

10 warning signs of primary immunodeficiency مهمة

1. • four or more ear infections in a year
2. • two or more severe sinus infections in a year
3. • two or more months of antibiotic treatment with little effect
4. • two or more pneumonias per year
5. • insufficient weight gain or growth delay
6. • recurrent deep skin or organ abscesses
7. • persistent mouth thrush or fungal skin infection
8. • need for iv antibiotics to clear infections
9. • two or more deep–seated infections
10. • family history of primary immunodeficiency

Investigation: according to suspected type of immunodeficiency:

1. T cell defect (cellular) :
   - CBC,differential ,Lymphocyte count
   - Subsets :CD4 – CD8 numbers
   - Candidal skin test
2. Ab B cell defect (Humoral) /
   - Ig level IgG – IgM – IgA – IgA
   - IgG subclasses (>2 year)
   - Specific Ab response (Vaccine included)
   - Lymphocyte subset
3. Combined B T cell defect :As above
4. Neutrophils defect :
   - CBC neutrophil count
   - Nitro blue tetrazolium test (NBT test)
   - Test for chemotaxis in (LAD)
5. Complement defect :CH50 – Complement level

Primary immune deficiencies

Clinical Presentation:

1. Recurrent bacterial infections
2. Sever infection (eg meningitis – pneumonia – osteomyelitis)
3. Extensive candidiasis
4. Infections that present atypically, are unusually severe or chronic or fail regular Rx
5. Infections caused by an unexpected or opportunists pathogen or child has been immunised against
6. Sever or long lasting warts, generalized molluscum contagiosum
7. Complication of vaccination (disseminated BGG)
8. Abscess of internal organs – recurrent sic Abscesses
9. Prolonged or recurrent diarrhea

Types

1. T cell Defects:

Criteria :Severe and/or unusual viral and fungal infections and faltering growth in first months of life,e.g. severe bronchiolitis, diarrhoea, oral thrush, Pneumocystis jirovecii (carinii) pneumonia (PCP), disseminated or severe cytomegalovirus infection.

- **Severe combined immunodeficiency (SCID): B&Tcell defect**
  - XLR
  - Defect in cellular B and humoral T.
  - Severe recurrent bacterial infection –viral–candida,FTT,
  - Fatal with out Rx (BMT or HSCT)
- **Wiskott –Aldrich syndrome**
  - XLR
  - Traid of :1–Immundefiecncy (combined) + 2–Thrombocytopenia (also small plt) +3– Eczema .
  - Rx – IVIG
    – PLt transfusion
    –HSCT
- **Digeorge Syndrome :** AD (Microdeletion 22q11)
  CATCH22
  C: CHD – TOF, interrupted arch of aorta
  A: Abnormal Face, short philtrum, low height ear and WLQ
  T: Thymic hypoplasia (Tell defect)
  C: Cleft palate
  H : Hypoparathyroidism (Hypo Ca in neonatal period)
  22: Chromosome 22
- **Ataxia – Telangiectasia:**
  - Defect in DNA repair
  - AR
  - Increase risk of malignancy (lymphoma–brain tumor)
  - Combined B+T defect

Recurrent infection (sinopulmonary)



Cerebellar Ataxia ≃2y Telangiectasia(skin – eye)≃ 3–6y

- Duncan syndrome
  - EBV infection
  - Lymphoma

2. B cell Defects:

- **Criteria: In the 1st year 2 years(>6months…?),sever bacterial infection especially sinu pulmonary,ear – skin infection, diarrhea, FTT**

- **X– Linked Agammaglobulinemia (bruton disease)**
  - Defect in bruton tyrosine kinases (BTK) gene , failure of differention , No B cells (Increase Ig)
  - ♂ Boy
  - Onset : 9 – 12 months
  - C/P :recurrent severe sinopulmonary bacterial infections.
  - No LD ,No splenomegaly ,No tonsillar enlargement.
  - Low total Ig , IgA – IgG – IgM
  - Rx : IVIG/ month – Antibiotic

- **Common variable immune deficiency: (CVID) :acquired hypogammaglobinemia)**
  - 2ry to drugs as phenytion .
  - B cells lie ,less severe infection – High risk of autoimmune disorders and malignancy
  - AD
  - Age :Late childhood ,10 years

- **Hyper IgM syndrome**
  - XL immunodeficiency
  - B cells defect (defective switch IgM to IgG)
  - High IgM –Low IgG,
  - C/P: Boy with recurrent month ulcer – verruca vulgaris recurrent infection (bacterial +pneumocystic carinii)

- **Selective IgA deficiency :AD**
  - Is the Most common 1ry Immunedefeceincy
  - 1:500 (coucasians)
  - Low IgA, Ig other Ig Normal or high
  - C/P: – Asymptomatic /recurrent sinopulmonary or GIT infection,diarrhea –FTT.

- Associated with celiac – JRA – SLE –Down syndrome.
- Antibiotics.
- IVIG is CI .

3. Neutrophil defects:

- C/P:Recurrent bacterial infections and abscess invasive fungal infections – FTT – Granuloma
- Example – **Chronic granulomatous disease** – most are X–linked recessive, some autosomal recessive. Neutrophils fail to produce superoxide after phagocytosis of microorganisms.

4. Leukocyte function defects:

- Delayed separation of umbilical cord, delayed wound healing, chronic skin ulcers and deep–seated infections.
- **Leucocyte adhesion deficiency (LAD)** – deficiency of neutrophil surface adhesion molecules, CD18, CD11b, leads to inability of neutrophils to migrate to sites of infection/inflammation.

5. Complement defects:

C/P: Recurrent bacterial infection (encapsulated organism as N– meningitidis)also increase risk of Autoimmune SLE .

Management : needs MDT , according to type of immunedefeceincy

- **Antimicrobial prophylaxis:**
  - For T–cell and neutrophil defects,cotrimoxazole to prevent PCP and itraconazole or fluconazole to prevent other fungal infections
  - For B–cell defects – antibiotic prophylaxis (e.g. azithromycin) to prevent recurrent bacterial (e.g. chest, ear, sinus) infections
- **Antibiotic treatment:**
  - Prompt treatment of infections
  - Appropriate choice of antibiotics to cover likely organisms
  - Generally longer courses, with low threshold for intravenous therapy
- **Immunoglobulin replacement therapies** :
  - For children with antibody deficiency ,
  - Can be given intravenously, which may require central venous (Portacath or Hickman) line insertion, or subcutaneously
- **Bone marrow transplantation**
  - E.g. for severe combined immunodeficiency, chronic granulomatous disease
  - Can be matched sibling donor, matched unrelated donor, or haploidentical (parental) transplant
- **Gene therapy (mcq)**
  - Currently an evolving area and not widely available

References:

1. Nelson textbook of pediatrics 22th edition.
2. Illustrated textbook of paediatrics 6th edition.
3. Nelson essentials of pediatrics 9th edition.
4. www.uptodate.com
5. www.medscape.com

تنبيه :

هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة

لهذا لايجوز نسخ أو اقتباس أي جزء من المحتوى الا بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي